

Potential & Conservation of Energy

Building Models to Describe our World
Conservation, Potential and Potential Uses



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- Force from potential energy

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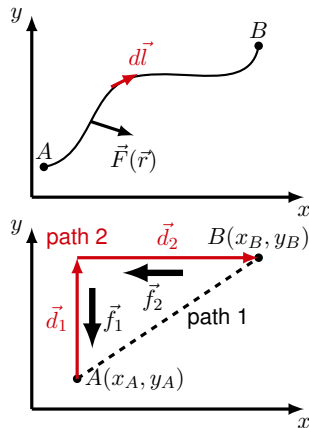
Conservative Forces

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Recall: Work

$$W = \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l}$$

- Force can change magnitude and direction based on position.
- Need to define work as a “path integral”, as it depends on the path taken.
- In general, this integral can be difficult to evaluate.



Work is a path integral.

Recall: Work-Energy Theorem

- The net work done on an object is its change in kinetic energy:

$$W^{net} = \Delta K$$

Recall: Work-Energy Theorem

Example: Object falling due to gravity

The work done by gravity:

$$W^{net} = \vec{F}_g \cdot \vec{d} = mgh$$

The change in kinetic energy (final - initial)

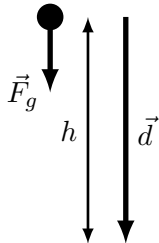
$$\Delta K = \frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$$

Work-Energy theorem:

$$W^{net} = \Delta K$$

$$mgh = \frac{1}{2}mv^2$$

$$\therefore v = \sqrt{2gh}$$



A mass, m , falls a distance, h , with a corresponding displacement, $\vec{d}$.

Conservative vs non-conservative forces

- A non-conservative force is one for which the work done depends on the path taken.
- A conservative force is one for which the work done does not depend on the path taken:
 - Corollary: The work done by a conservative force along a closed path is zero.

Condition for a force to be conservative

- The work done on a closed path must be zero:

$$W = \int_A^A \vec{F}(\vec{r}) \cdot d\vec{l} = \oint \vec{F}(\vec{r}) \cdot d\vec{l} = 0$$

- This requires the following conditions on, $\vec{F}(\vec{r})$, (given by Stokes' Theorem, which is way beyond the scope of this course):

$$\begin{aligned}\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} &= 0 \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} &= 0 \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} &= 0\end{aligned}$$

... evaluating these partial derivatives is however not beyond the scope of this course!

Example of a conservative force

- The force from a spring:

$$\vec{F}(\vec{r}) = \vec{F}(x, y, z) = F_x \hat{x} + F_y \hat{y} + F_z \hat{z} = -kx \hat{x} + 0\hat{y} + 0\hat{z}$$

- Take one of the terms:

$$\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = \frac{\partial}{\partial z}(-kx) - \frac{\partial}{\partial x}0 = 0$$

- They are all zero $\rightarrow$ The force from a spring is conservative !

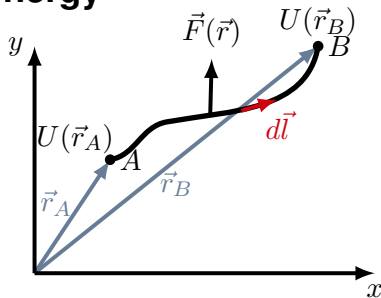
Work by a conservative force – Potential energy

- The work done by a conservative force does not depend on the path, only on the end points:

$$W = \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l}$$

- We introduce a **function of position**, $U(\vec{r})$, such that the **negative of the work** done by the conservative force can be calculated using that function:

$$-W = - \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l} = U(\vec{r}_B) - U(\vec{r}_A) = \Delta U$$



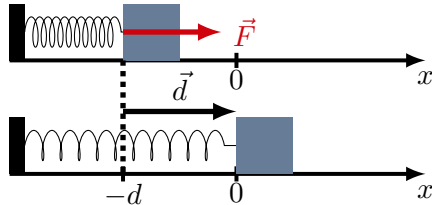
Work done by a conservative force depends only on the end points.

$U(\vec{r})$ is called the **potential energy function**. It is associated with a specific conservative force and **depends only on position**.

Example: Potential energy for a spring

- We want to know the potential energy function for a spring. We start with Work:

$$-W = - \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l} = U(\vec{r}_B) - U(\vec{r}_A)$$



Work done by a spring force, $\vec{F}(x) = -kx\hat{x}$,
from $x = -d$ to $x = 0$.

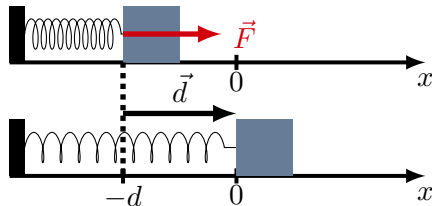
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$$-W = - \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l} = U(\vec{r}_B) - U(\vec{r}_A)$$

- Work from the spring in going from A ($x = -d$) to B ($x = 0$):

$$-W = - \int_{-d}^0 -kx\hat{x} \cdot d\mathbf{x}\hat{x} = \int_{-d}^0 kx dx = \frac{1}{2}k0^2 - \frac{1}{2}kd^2$$



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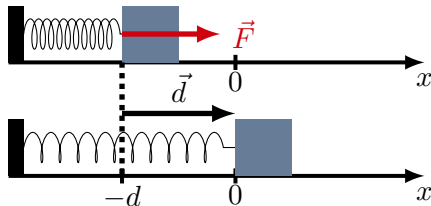
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$$-W = - \int_{-d}^0 -kx\hat{x} \cdot d\vec{x} = \int_{-d}^0 kx dx = \frac{1}{2}k0^2 - \frac{1}{2}kd^2$$

- Identify:

$$U(x) = \frac{1}{2}kx^2 + C$$



Work done by a spring force, $\vec{F}(x) = -kx\hat{x}$, from $x = -d$ to $x = 0$.

We can only determine potential energy up to a constant, since **only the difference in $U(x)$ is related to work**. Like an anti-derivative...

Work-Energy Theorem and potential energy

- The net work done can be split into the sum of work done by conservative and non-conservative forces:

$$W^{net} = W^{NC} + W^C = \Delta K$$

- Re-arrange:

$$W^{NC} = \Delta K - W^C$$

- Use potential energy for work done by conservative force:

$$W^{NC} = \Delta K + \Delta U$$

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- Use potential energy for work done by conservative force:

$$W^{NC} = \Delta K + \Delta U$$

- This is why we introduced a minus sign when defining change in potential energy as the negative of work done, $-W = \Delta U$.

Interpretation

- The work done by non-conservative forces will change the “energy” (potential + kinetic) in the system:

$$W^{NC} = \Delta K + \Delta U = \Delta E$$

- Work done by non-conservative forces can *add or remove* energy from the system:
 - You push on a crate (do positive work), that energy goes into kinetic energy of the crate and/or (say) gravitational potential energy (if you push the crate up an incline).
 - A crate slides down an incline, the (negative) work done by friction removes energy, so the crate will have less kinetic energy than it otherwise would have.

Force from potential energy (e.g. for a spring)

- Effectively, the **potential energy can be found from the anti-derivative** of a force function (if we take the path integral on a path parallel to the direction of the force):

$$-W = - \int_{-d}^0 F(x)dx = \int_{-d}^0 kx dx = \frac{1}{2}k0^2 - \frac{1}{2}kd^2 = U(0) - U(-d)$$

where we easily identified $U(x) = \frac{1}{2}kx^2 + C$.

- The force can then be found from the derivative of the potential energy function:

$$F(x) = -\frac{dU}{dx} = -\frac{d}{dx} \left(\frac{1}{2}kx^2 + C \right) = -kx$$

The signs matter, be careful!

Force from potential energy in 3D

- In 3D, the potential energy is a **scalar function** of space, $U(x, y, z)$.
- The force **vector** is given by the negative of the **gradient vector** of the potential energy function (which you find with the partial derivatives):

$$\vec{F} = -\nabla U(x, y, z)$$

$$F_x = -\frac{\partial}{\partial x}U(x, y, z)$$

$$F_y = -\frac{\partial}{\partial y}U(x, y, z)$$

$$F_z = -\frac{\partial}{\partial z}U(x, y, z)$$

Conservation of Energy

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Potential energy - recap

- Some forces are conservative:
 - Can define a potential energy for those forces.
 - Potential energy is only defined up to a constant (that you can choose).
- **Change in potential energy** is equal to the negative of the work done by that force:

$$\Delta U = -W = - \int_A^B \vec{F}(\vec{r}) \cdot d\vec{l}$$

- Can use this in the work-energy theorem:

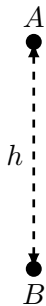
$$W^{net} = W^{NC} + W^C = \Delta K$$
$$\therefore W^{NC} = \Delta K + \Delta U$$

Potential energy

- Potential energy depends on position (and thus on an arbitrary choice of coordinate system).
- Only **changes** in potential energy are meaningful:
 - *Change* in potential energy does not depend on our choice of coordinate system.
 - Net work done is change in kinetic energy.
 - Work done by a conservative force is change in associated potential energy.
 - If you think about potential being the anti-derivative of force, then only difference in anti-derivatives (aka definite integrals) give useful physical information.
- Potential energy increases when you move in the direction opposite of the associated force (think gravity, mgh).

Changes in energy are independent of choice of coordinates

Consider a ball falling a total distance, h , from A to B . Is the change in potential energy positive or negative?



A mass, m , falls a distance, h .

Changes in energy are independent of choice of coordinates

Consider a ball falling a total distance, h , from A to B . Is the change in potential energy positive or negative?

$$U(x) = mgx$$

$$U_A = mgx_A$$

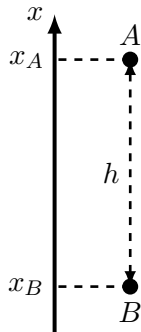
$$U_B = mgx_B$$

$$\Delta U = U_B - U_A$$

$$= mgx_B - mgx_A$$

$$= mg(x_B - x_A)$$

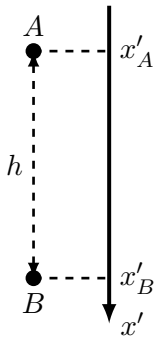
$$\therefore \Delta U = -mgh$$



A mass, m , falls a distance, h .

Changes in energy are independent of choice of coordinates

Consider a ball falling a total distance, h , from A to B . Is the change in potential energy positive or negative?



A mass, m , falls a distance, h .

$$U(x') = -mgx'$$

$$U_A = -mgx'_A$$

$$U_B = -mgx'_B$$

$$\Delta U = U_B - U_A$$

$$= -mgx'_B - (-mgx'_A)$$

$$= -mg(x'_B - x'_A)$$

$$\therefore \Delta U = -mgh$$

Mechanical energy

- We can define the mechanical energy of an object:

$$E = K + U$$

- If the object went from A to B :

$$E_A = K_A + U_A$$

$$E_B = K_B + U_B$$

- The change in mechanical energy is:

$$\Delta E = E_B - E_A = (K_B + U_B) - (K_A + U_A)$$

$$\therefore \Delta E = \Delta K + \Delta U$$

- The work done by non-conservative force is thus equal to the change in mechanical energy of the object:

$$W^{NC} = \Delta E = \Delta K + \Delta U$$

Interpretation – Conservation of energy

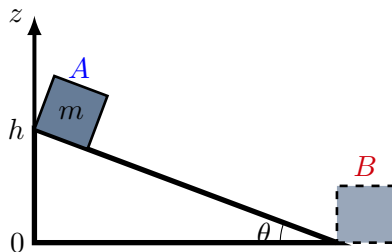
$$W^{NC} = \Delta E = \Delta K + \Delta U$$

- An object can have potential and/or kinetic energy.
- Non-conservative forces can do work and change that energy:
 - Positive work $\rightarrow$ The energy increases.
 - Negative work $\rightarrow$ The energy decreases.
 - No work $\rightarrow$ The energy is constant.
- We call this “conservation of (mechanical) energy”.
- This is the origin of the term “conservative” for a force. It corresponds to those forces for which a potential energy can be defined and results in the total energy being conserved.

Conservation of energy

- We can model friction as doing negative work on the block, thus decreasing the mechanical energy of an object:

$$W^{NC} = E_B - E_A$$

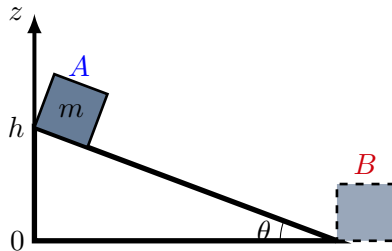


A block slides down an incline with friction.

Conservation of energy

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$$W^{NC} = E_B - E_A$$



A block slides down an incline with friction.

$$U(z) = mgz$$

$$E_A = mgh \text{ and } E_B = \frac{1}{2}mv^2$$

$$W^{NC} = -f_k \frac{h}{\sin \theta}$$

$$\therefore -f_k \frac{h}{\sin \theta} = \frac{1}{2}mv^2 - mgh$$

Conservation of energy

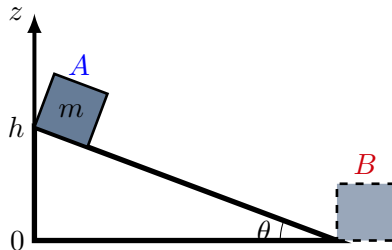
- We can model friction as doing negative work on the block, thus decreasing the mechanical energy of an object:

$$W^{NC} = E_B - E_A$$

- We can think of this as:

$$E_B = E_A + W^{NC}$$

- The final energy (E_B) is equal to the initial energy (E_A) plus(minus) any energy added(removed), W^{NC} , to(from) the system by non-conservative forces.



A block slides down an incline with friction.

$$U(z) = mgz$$

$$E_A = mgh \text{ and } E_B = \frac{1}{2}mv^2$$

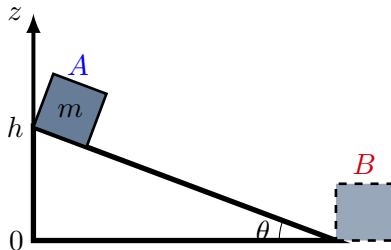
$$W^{NC} = -f_k \frac{h}{\sin \theta}$$

$$\therefore -f_k \frac{h}{\sin \theta} = \frac{1}{2}mv^2 - mgh$$

$$\frac{1}{2}mv^2 = mgh - f_k \frac{h}{\sin \theta}$$

Conservation of energy

- We can think of friction as increasing the “thermal energy” of the block (it heats up), Q , and include an additional type of energy.
- In general we can think of W^{NC} as some external agent injecting or removing energy, it doesn't have to be mechanical work and we can always account for it in terms of an energy.
- If we include thermal energy in the system to model friction:
 - All remaining forces are conservative.
 - Energy is always conserved.
 - We should include the incline into our “system” since it will share the heat with the block.



A block slides down an incline with friction.

$$Q_B = f_k \frac{h}{\sin \theta}$$

$$E_A = mgh$$

$$E_B = \frac{1}{2}mv^2 + Q_B$$

$$W^{NC} = 0 \rightarrow E_B = E_A$$

$$\therefore \frac{1}{2}mv^2 + f_k \frac{h}{\sin \theta} = mgh$$

Modeling Potential

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Conservation of mechanical energy

- When there is no net work being done by non-conservative forces, the mechanical energy is constant:

$$E = \text{const.} = E_A = E_B$$
$$\Delta E = 0 = \Delta U + \Delta K$$

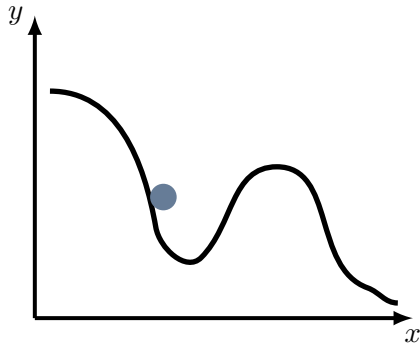
- Potential and Kinetic energy “trade off” with each other:

$$\Delta K = -\Delta U$$

- If the potential energy increases, the kinetic energy must decrease by the same amount, and vice versa, *just like the pendulum*.

Example: marble on a track

- If I place a marble on a track, it will move under the influence of gravity.
- If I place it higher, it will have more initial energy, E .
- The height at which I place it is a measure of its total energy, E (it's all potential energy when I place the marble at rest).
- As the marble rolls along the track, it will trade off kinetic and potential energy.



A marble on a track in the vertical-horizontal plane.

Example: marble on a track

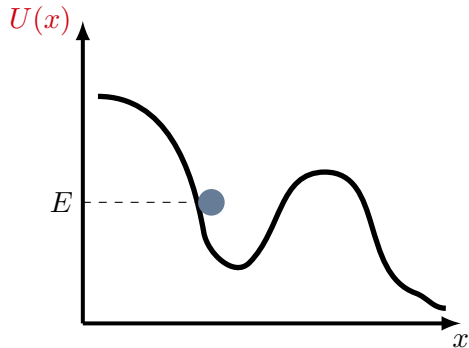
- The track **is** the gravitational potential energy as a function of horizontal position:

$$U(x) = mgy(x) \propto y(x)$$

- The initial “height” of the marble is the energy of the marble.
- A potential energy function is like a track!
- The energy of the marble is constant:

$$E = U(x) + K$$

$$K = E - U(x)$$

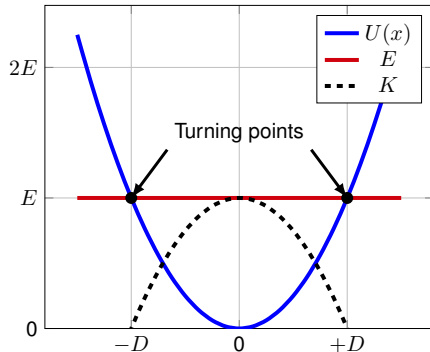
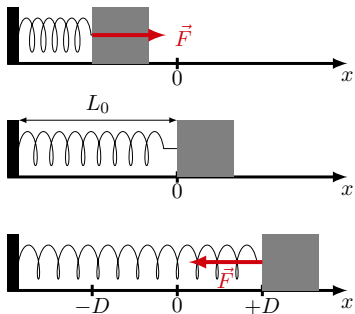


In 1D, the potential energy function can be thought of as a track for a marble.

Potential energy for a spring

We can interpret any potential energy function as if it were a track upon which we put a marble.

$$U(x) = \frac{1}{2}kx^2$$



Modeling with energy (how to)

- The most important step is ~~identifying the forces~~ identifying the potential energies and sources of non-conservative work.
- Compare energy at some point (time) A and some other point (time) B; the difference in mechanical energies is the work done by non-conservative forces:

$$E_A = K_A + U_{1A} + U_{2A} + \dots$$

$$E_B = K_B + U_{1B} + U_{2B} + \dots$$

$$\rightarrow W^{NC} = E_B - E_A$$

The Lagrangian: A completely different approach

- You won't be tested on your knowledge of this, however...
- It is presented to you as a different way to build models:
 - We give you a prescription and tell you that if you follow those rules, you will develop a model that describes the world.
 - You should be open to that process (following rules, not understanding the rules – nobody can seriously claim to fully understand them).
- The Lagrangian method is very powerful, carries into Quantum Mechanics, and is mathematically very aesthetic. It almost feels as if there is an underlying beauty to it all...

The prescription

1. Build the Lagrangian:

$$L = K - U$$

2. Apply the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial v_x} \right) - \frac{\partial L}{\partial x} = 0$$

This gives the equations of motion (e.g. the acceleration).

You get one such equation per coordinate (e.g. replace (x, v_x) with (y, v_y) if these also appear in L).

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You get one such equation per coordinate (e.g. replace (x, v_x) with (y, v_y) if these also appear in L).

For a particle falling with gravity (x positive upwards):

$$L = \frac{1}{2}mv^2 - mgx$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial v_x} \right) = \frac{d}{dt} (mv) = ma$$

$$\frac{\partial L}{\partial x} = -mg$$

$$\therefore ma + mg = 0$$

$$a = -g$$

As expected, the acceleration is $-g$, since x is positive upwards.

Lagrangian for a pendulum

For a pendulum, we use the angle θ as the coordinate to describe position. The speed is given by $v = \omega L$, where $\omega = \frac{d\theta}{dt}$:

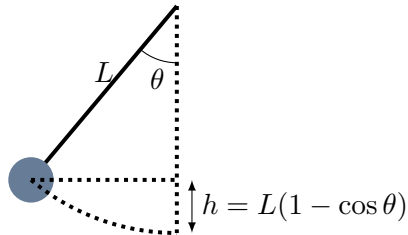
$$K = \frac{1}{2}mv^2 = \frac{1}{2}mL^2\omega^2$$

$$U = mgh = mgL(1 - \cos \theta)$$

$$\therefore L = K - U = \frac{1}{2}mL^2\omega^2 - mgL(1 - \cos \theta)$$

The Euler-Lagrange equation is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \omega} \right) - \frac{\partial L}{\partial \theta} = 0$$



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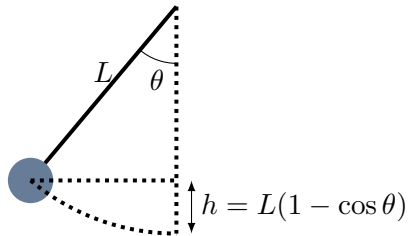
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$$\therefore L = K - U = \frac{1}{2}mL^2\omega^2 - mgL(1 - \cos \theta)$$

The Euler-Lagrange equation is:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \omega} \right) - \frac{\partial L}{\partial \theta} = 0$$



$$\frac{d}{dt} \left(\frac{\partial L}{\partial \omega} \right) = \frac{d}{dt} (mL^2\omega) = mL^2\alpha$$

$$\frac{\partial L}{\partial \theta} = -mgL \sin \theta$$

$$\therefore mL^2\alpha + mgL \sin \theta = 0$$

$$\alpha = \frac{g}{L} \sin \theta$$

Emmy Noether's theorem

- If the Lagrangian is “invariant to some symmetry”, then something is conserved.
- This ties the idea of a conserved quantity to some symmetry in the laws of nature.
- For example, if the Lagrangian does not change with time, then energy is conserved.
- We now see conservation laws as the result of symmetries in the laws of physics.
- Momentum, charge, energy are conserved because the laws of physics are invariant to different “symmetries”.



Emmy Noether. (2025, August 12). Wikipedia, the free encyclopedia. Retrieved August 25, 2025, from https://en.wikipedia.org/wiki/Emmy_Noether

The Standard Model of particle physics

- The Standard Model of Particle physics describes the three forces (weak, strong, e/m) of nature that are not gravity. It's just a Lagrangian.
- Each term represents some interaction between particles.
- Only certain functional forms are allowed in order to preserve symmetries.
- So far, it appears that every (mathematically) allowable term describes something in nature. That's pretty cool.
- People have discovered particles based on the idea that a particular term could be present in the Lagrangian.

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{8}\text{tr}(\mathbf{W}_{\mu\nu}\mathbf{W}^{\mu\nu}) - \frac{1}{2}\text{tr}(\mathbf{G}_{\mu\nu}\mathbf{G}^{\mu\nu}) \\ & + (\bar{\nu}_L, \bar{e}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} + \bar{e}_R \sigma^\mu i D_\mu e_R + \bar{\nu}_R \sigma^\mu i D_\mu \nu_R + (\text{h.c.}) \\ & - \frac{\sqrt{2}}{v} \left[(\bar{\nu}_L, \bar{e}_L) \phi M^e e_R + \bar{e}_R \tilde{M}^e \tilde{\phi} \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} \right] \\ & - \frac{\sqrt{2}}{v} \left[(-\bar{e}_L, \bar{\nu}_L) \phi^* M^\nu \nu_R + \bar{\nu}_R \tilde{M}^\nu \phi^T \begin{pmatrix} -e_L \\ \nu_L \end{pmatrix} \right] \\ & + (\bar{u}_L, \bar{d}_L) \tilde{\sigma}^\mu i D_\mu \begin{pmatrix} u_L \\ d_L \end{pmatrix} + \bar{u}_R \sigma^\mu i D_\mu u_R + \bar{d}_R \sigma^\mu i D_\mu d_R + (\text{h.c.}) \\ & - \frac{\sqrt{2}}{v} \left[(\bar{u}_L, \bar{d}_L) \phi M^d d_R + \bar{d}_R \tilde{M}^d \tilde{\phi} \begin{pmatrix} u_L \\ d_L \end{pmatrix} \right] \\ & - \frac{\sqrt{2}}{v} \left[(-\bar{d}_L, \bar{u}_L) \phi^* M^u u_R + \bar{u}_R \tilde{M}^u \phi^T \begin{pmatrix} -d_L \\ u_L \end{pmatrix} \right] \\ & + (\overline{D_\mu \phi}) D^\mu \phi - m_h^2 [\tilde{\phi} \tilde{\phi} - v^2/2]^2/2v^2.\end{aligned}$$

The Lagrangian for the Standard Model.